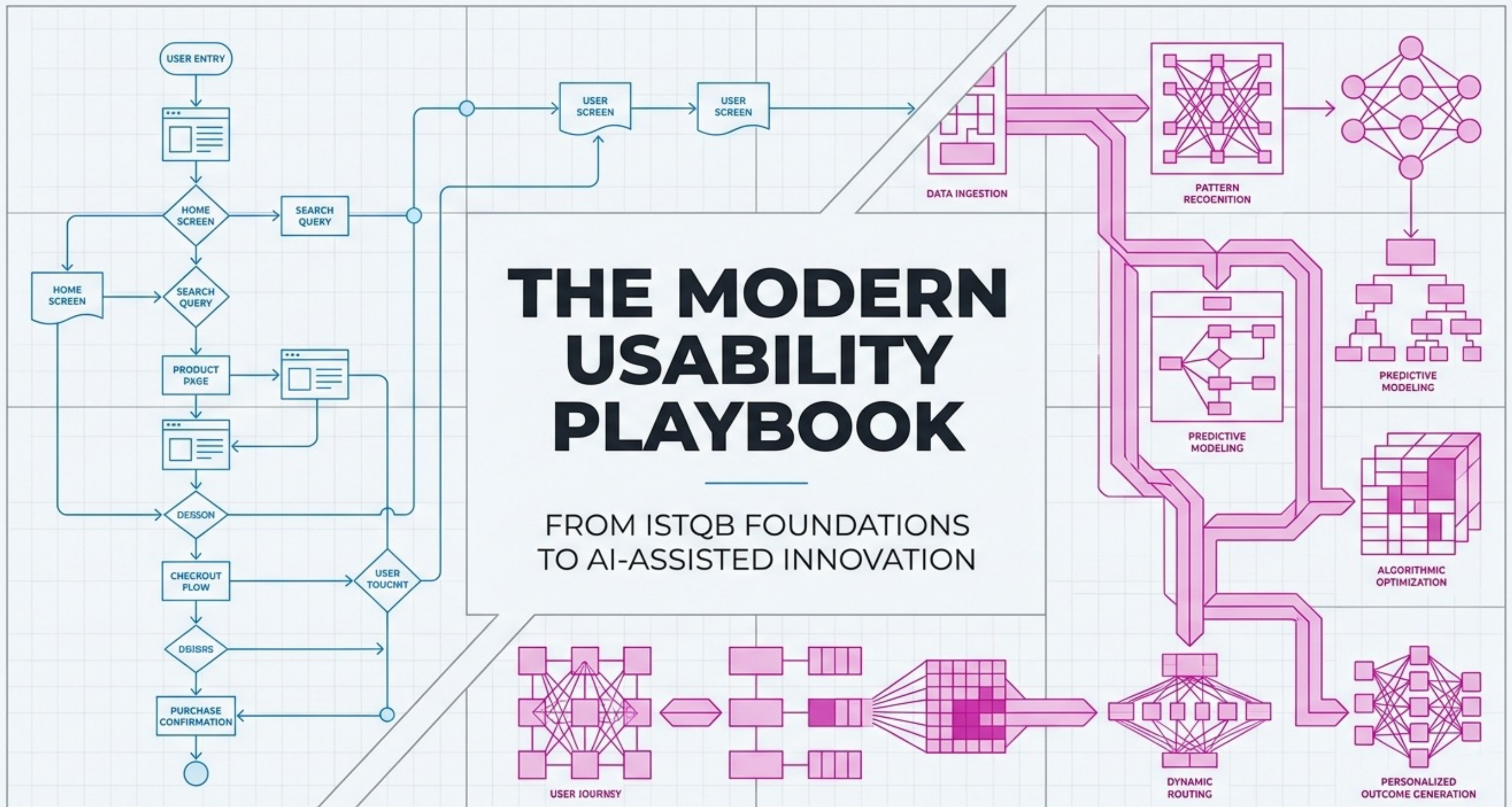




The UX Landscape

Layer 3: User Experience (UX)

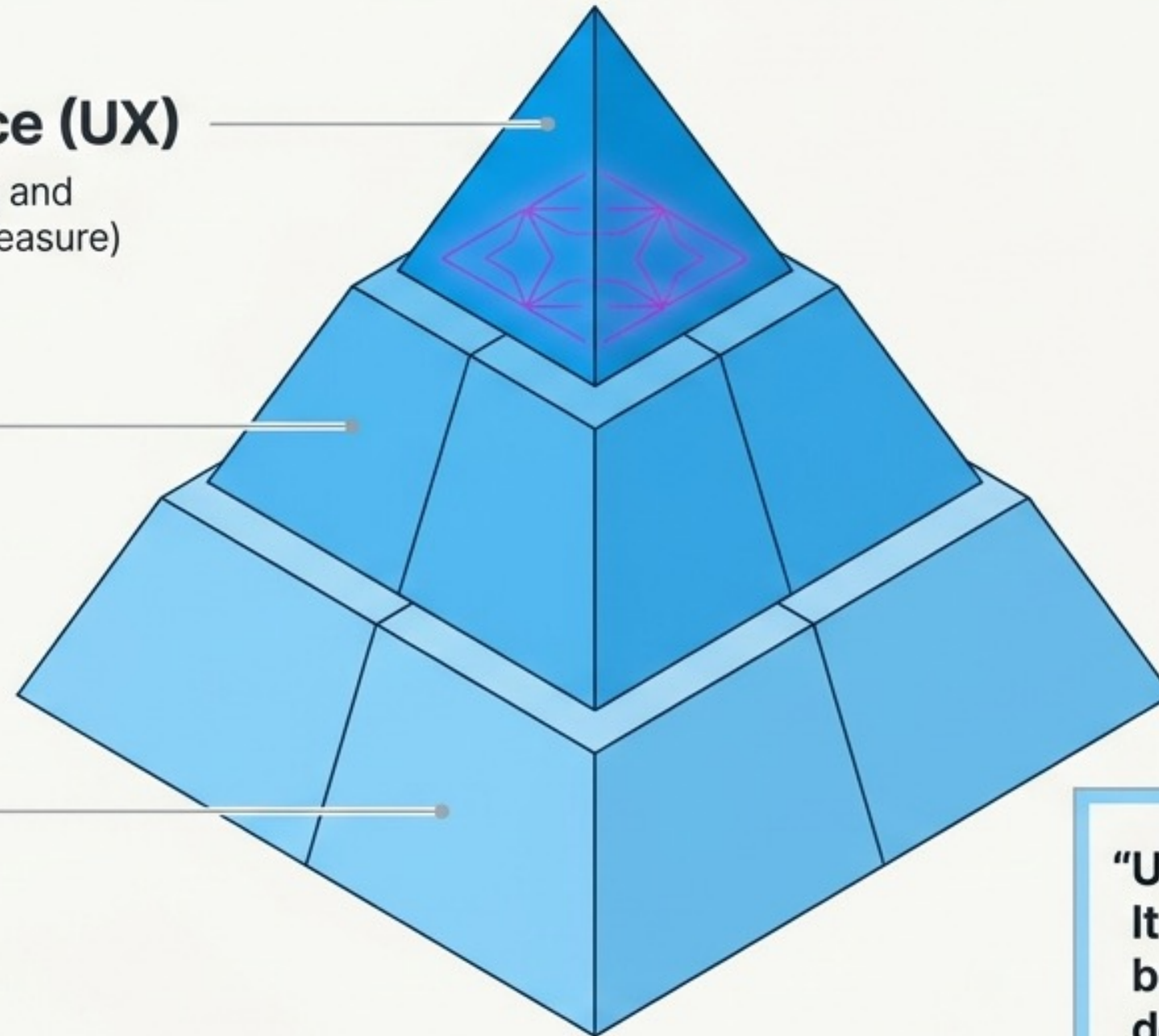
Emotional: How do people feel during and after the interaction? (Satisfaction, Pleasure)

Layer 2: Usability

Operational: How well can users actually utilize that functionality? (Effectiveness, Efficiency, Safety)

Layer 1: Use / Utility

Functional: Can the system do what is needed to achieve a goal?



“Usability is not a single dimension. It is the critical, measurable bridge between raw utility and a delightful user experience.”

The 5 Pillars of Usability

Learnability

The steepness of the novice learning curve. True 'walk up and use' systems aim for zero learning time.

Efficiency

The speed and resource utilization of an expert user completing specified tasks.

Memorability

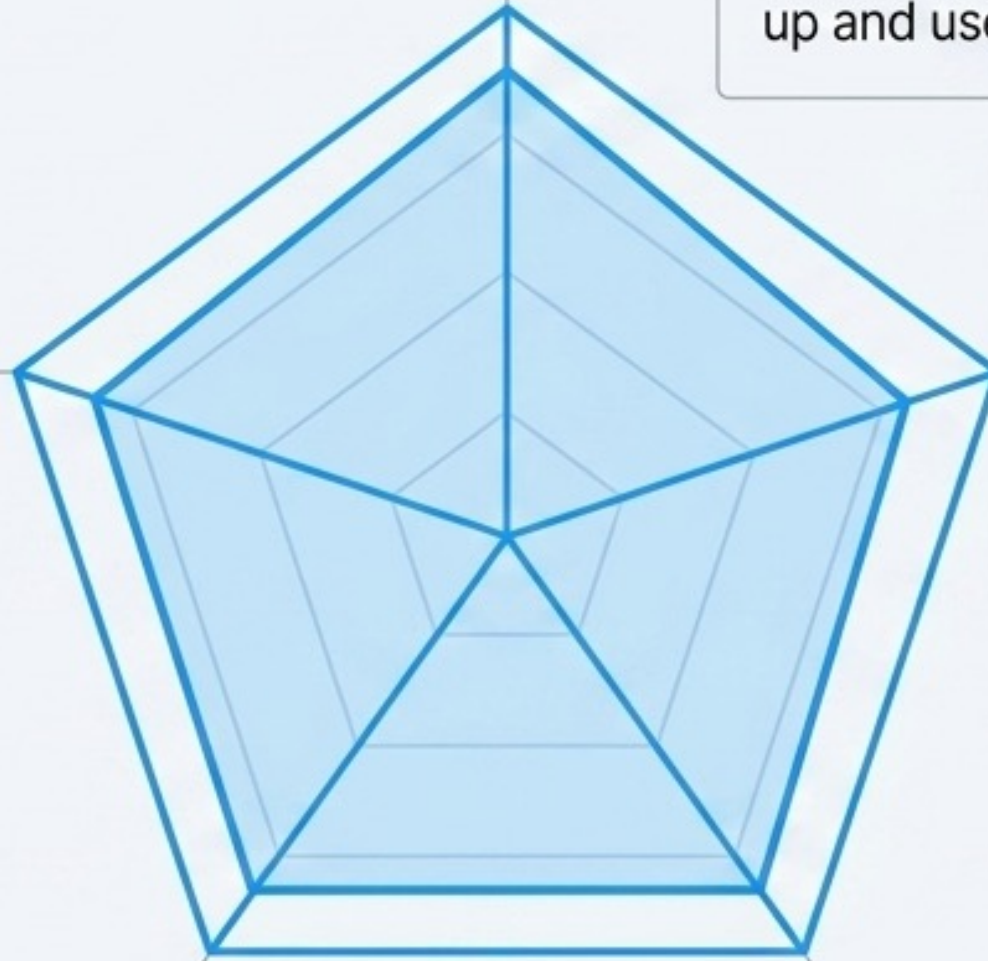
The friction experienced by an intermittent user returning to the system after a period of absence.

Satisfaction

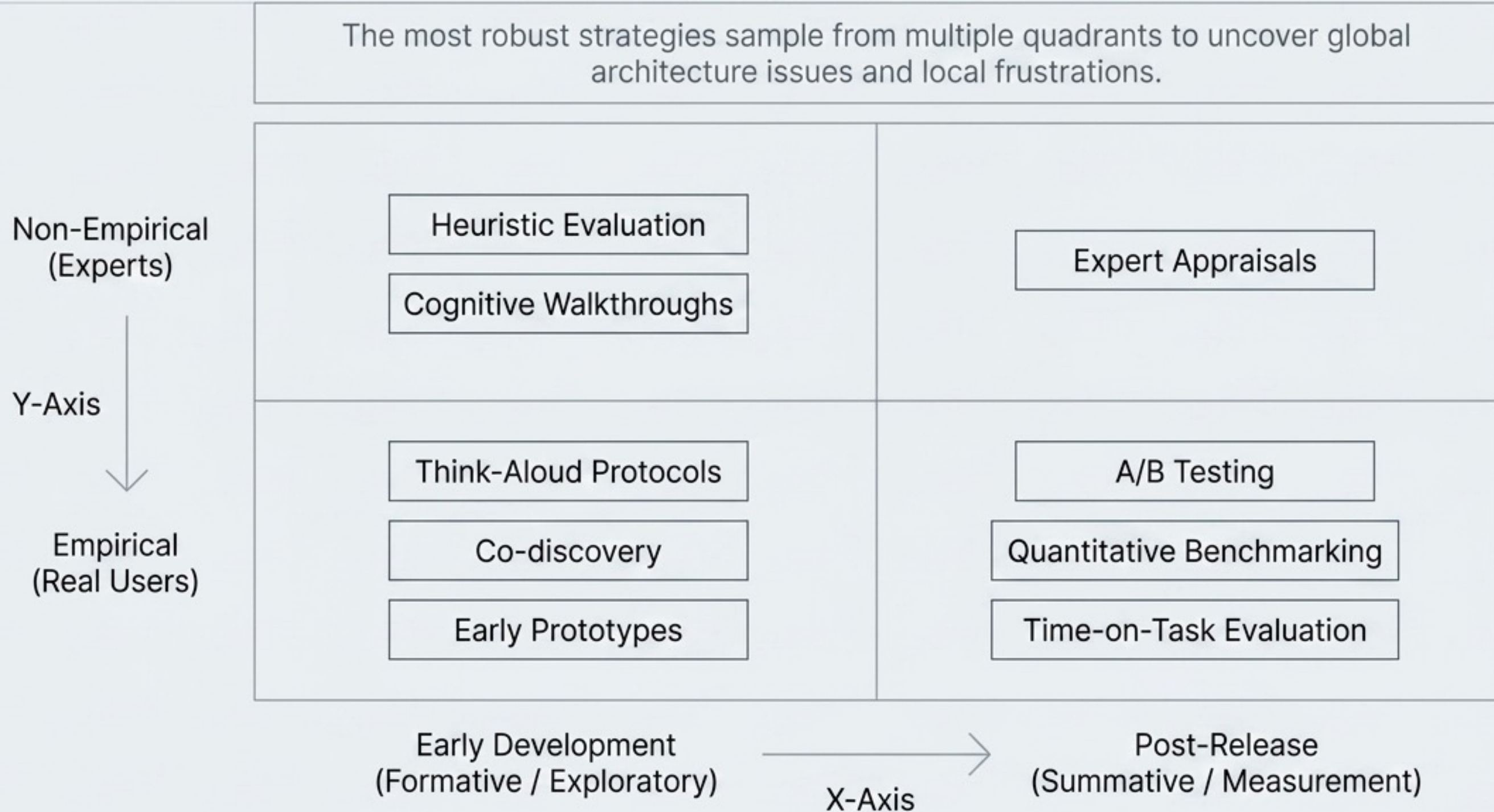
The subjective comfort and acceptability of the system as a means to achieve the user's goals.

Errors (Safety)

The frequency and catastrophic impact of user mistakes.



The Evaluation Strategy Matrix



The Usability Testing Lifecycle



1. Preparation

Define specific scenarios, isolate variables, and select a representative user sample.



2. Introduction

Establish psychological safety. Emphasize that the system is being tested, not the user.



3. Execution

Unbiased observation. The test leader maintains absolute neutrality with zero interference.



4. Debriefing

Collect immediate subjective feedback (e.g., questionnaires) before discussion introduces bias.

Translating Goals to Metrics

	Goal	Concern (Question)	Metric
Effectiveness	Users select correct navigation paths.	Are they clicking the right menu?	Count of wrong menu choices per session.
Efficiency	Minimize task duration.	Are they solving tasks quickly?	Time on task (measured in seconds).
Satisfaction	Users trust and enjoy the interface.	Are they comfortable and confident?	System Usability Scale (SUS) score.

Collect only what maps directly to your predefined concerns. Testing everything tests nothing.

What is SUS Score?

SUS (System Usability Scale) is a classic standardized questionnaire used to quantify the perceived usability of a product or system. In simple terms, it uses a **score from 0 to 100** to measure how usable users perceive a system (such as a website, app, or a kiosk like Omninet AI). The score represents the user's subjective experience, rather than identifying specific interface problems.

SUS Questionnaire Content

The SUS questionnaire consists of just 10 concise questions. Users rate each statement from 1 (strongly disagree) to 5 (strongly agree) based on their feelings.

The items are carefully designed: odd-numbered items are generally positive statements, while even-numbered items are negative statements, allowing cross-validation of user responses.

The SUS score is a reliable metric for measuring user satisfaction and system usability. It typically correlates with usability risks defined in usability test reports and risk analyses, such as those for the Omninet AI project.

- | No. | Question |
|-----|---|
| 1 | I think that I would like to use this system frequently. |
| 2 | I found the system unnecessarily complex . |
| 3 | I thought the system was easy to use . |
| 4 | I think that I would need the support of a technical person to be able to use this system. |
| 5 | I found the various functions in this system were well integrated . |
| 6 | I thought there was too much inconsistency in this system. |
| 7 | I would imagine that most people would learn to use this system very quickly . |
| 8 | I found the system very cumbersome to use. |
| 9 | I felt very confident using the system. |
| 10 | I needed to learn a lot of things before I could get going with this system. |

Interpreting SUS Scores

SUS scores are calculated using a standard formula (scores are transformed and then summed). The interpretation is as follows:

SUS Score	Grade	Interpretation
< 51	F (Fail)	Poor
51 – 68	D (Marginal)	Below average
68 – 73	C (OK)	Average
73 – 79.9	B (Good)	Above average
80 – 100	A (Excellent)	Best possible

Reporting for Impact: Adapting to the Stakeholder

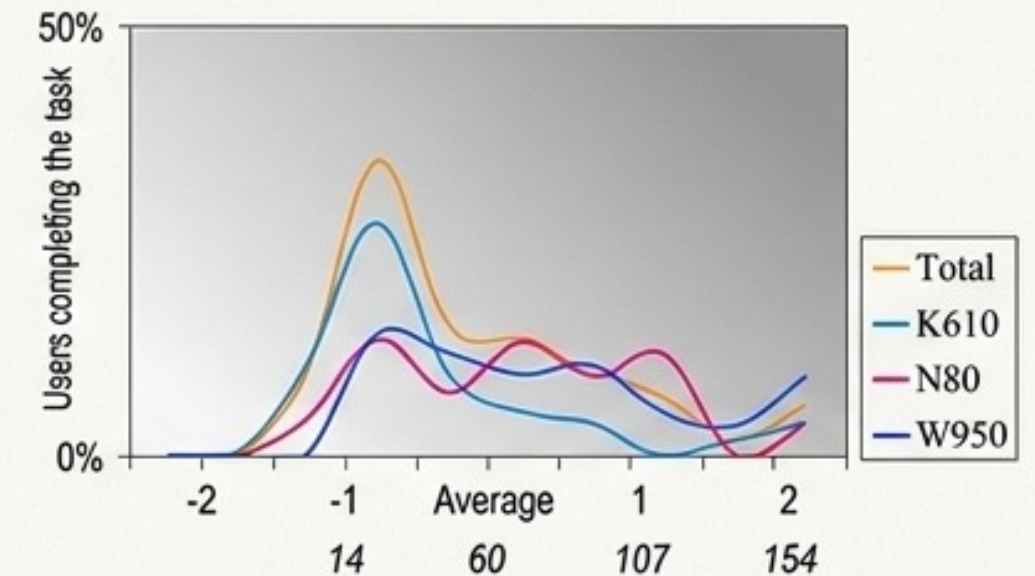
The Designer View

SDS matrices map qualitative complaints to specific UI blockers.

User	1	2	3	4	5	6
UserID	M1:1	F1:1	F3:1	F2:1	F1:2	M2:1
W=XXX, N=YYY, K=ZZZ	K:N:W	N:W:K	K:W:N	W:K:N	N:K:W	W:N:K
XXX						
UC1 - Receive and Answer an incoming call						
<i>Didn't know where to press</i>						
<i>Hesitated to press on screen</i>	y	y				
<i>Tried to press HW under screen</i>					y	
<i>Tried to press where says 'Answer?'</i>						
UC2 - Save the incoming call as a new contact - "Joanne"						
<i>Save Dialog</i>	y		y			y
<i>Name in call log doesn't update after saving number</i>				y		
UC3 - Set an alarm for 8 o'clock tomorrow morning						
<i>Couldn't find alarms</i>	y	y	y	n		y
<i>Puzzled by time view - where alarms</i>						
UC4 - Read an incoming SMS and reply with "I'm fine"						
UC5 - Make a phone call to Joanne(07949 877249)						
<i>Looking for Green and Red</i>						
UC6 - Create a new SMS - "Hi meet at 5" and send to Joanne						
<i>New SMS flow confusing - To: first</i>	y					y
<i>How enter contact - when To: in focus - no "contacts"</i>	y					
<i>Thought space on zero</i>						

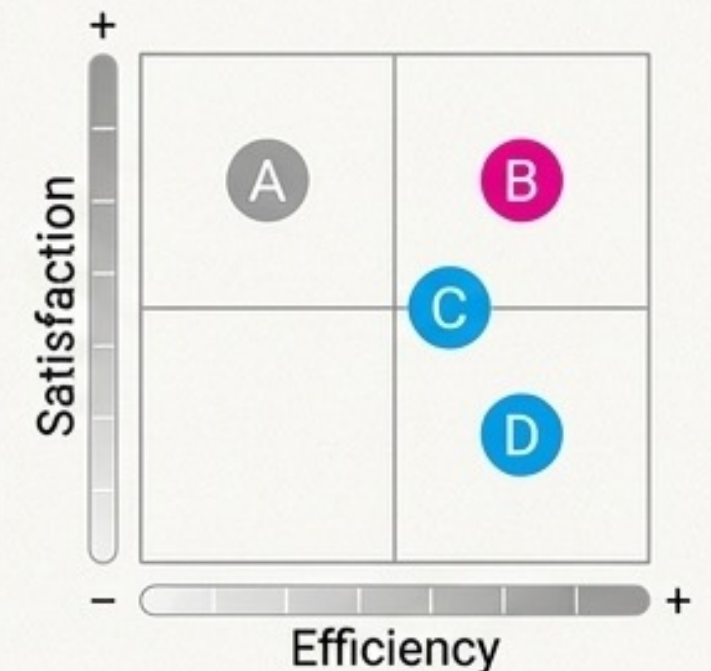
The Engineer View

Bell curves revealing use-case completion times and variance.



The Product Owner View

Spatial grids plotting overall satisfaction against efficiency.



The Bottleneck: Why Usability Needs an Upgrade

The Traditional Value

Deep **human empathy**

Ground-truth behavioral data

Uncovering **"unknown unknowns"**

The Constraints

Expensive, slow participant recruitment

Dozens of hours required for transcript synthesis

Subjective bias in manual evaluations

**AI does not replace the human user.
It accelerates the analytical
machinery surrounding them.**

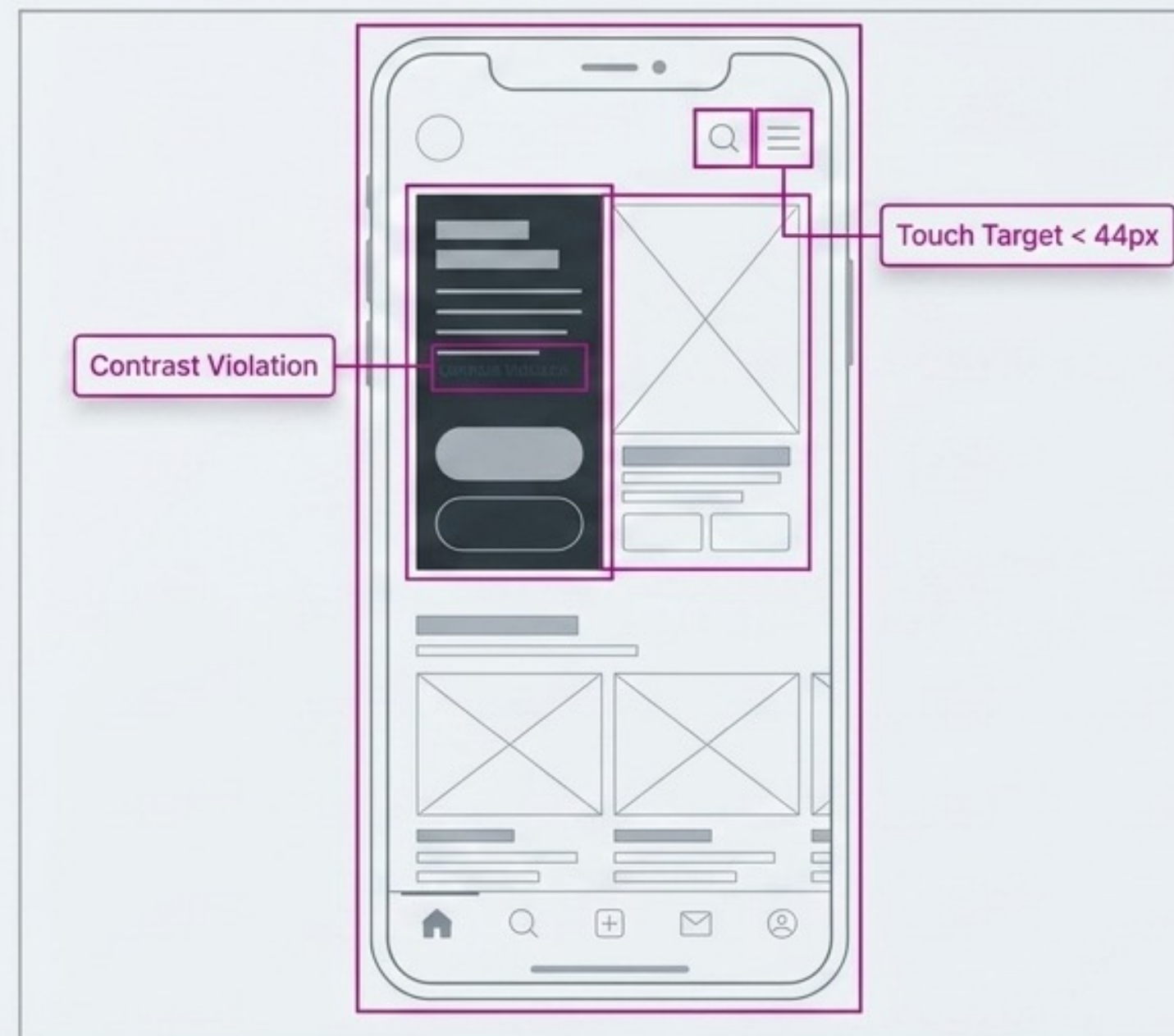
Case Study 1: Automated Heuristics

The Traditional Approach

3-5 UX experts spend days manually evaluating an interface against Nielsen's heuristics to find basic flaws.

The AI Acceleration

Multimodal LLMs scan Figma screens to instantly flag **contrast** issues, inconsistent navigation, and standard heuristic violations before a human ever sees it. Eliminates basic structural flaws in seconds.



Nielsen's heuristics

No.	Principle	Core Meaning
1	Visibility of system status	The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.
2	Match between system and the real world	The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system-oriented terms.
3	User control and freedom	Users need to be given full control over the system, especially the ability to undo and redo erroneous actions.
4	Consistency and standards	Within the same platform or product family, the same words, functions and operations should be consistent and not cause confusion.
5	Error prevention	The best design prevents problems before they occur, or provides optional confirmation dialogs.
6	Recognition rather than recall	Make options, actions, etc. visible so that the user does not have to remember information – “out of sight, out of mind” is avoided.
7	Flexibility and efficiency of use	Design for the majority of users (intermediate) while also accommodating beginners and experts.
8	Aesthetic and minimalist design	Interfaces should not contain irrelevant or rarely needed information; they should highlight the essentials and remove distractions.
9	Help users recognize, diagnose, and recover from errors	Error messages should clearly indicate the problem in plain language and suggest a constructive solution.
10	Help and documentation	Even if the system is designed well, it is necessary to provide easy-to-find, focused help documentation.

How to Perform a Heuristic Evaluation

Conducting a heuristic evaluation generally follows these steps:

- 1. Form an evaluation team** – Nielsen recommends **3-5 evaluators**. The team can include interaction designers, product managers, developers, etc., to ensure multiple perspectives.
- 2. Train the evaluators** – Make sure each member is familiar with the 10 heuristics, the scope of the evaluation, and the target user profiles.
- 3. Evaluate independently** – Each evaluator simulates user tasks, inspects the interface against the principles, and records findings on an evaluation sheet.
- 4. Aggregate and discuss** – Collect all problem reports, discuss them in a meeting to confirm which issues are real and prioritise them, then schedule fixes. The whole process can often be completed in a few hours.

[“Usability Omninet AI Heuristic Report”](#)

Case Study 2: Synthetic User Personas

Avatar



Persona: Elderly User, Tech-Averse

Goal: Find the accessibility menu.

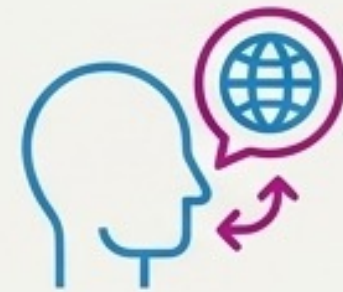
Avatar



Persona: Power User, Time-Poor

Goal: Bulk-edit data entries.

Avatar



Persona: Non-Native Speaker

Goal: Navigate onboarding flow.

The Process

LLMs are prompted with distinct cognitive profiles to **simulate rapid Cognitive Walkthroughs** via text or **UI-component trees**.

The Value Prop

Predicts friction points for niche demographics **instantly, refining** the test script to a **higher baseline** before deploying **expensive human participants**.

Case Study 3: Intelligent Synthesis & Debrief



Input: Raw think-aloud recordings & transcripts



The AI Action: Automatic transcription, timestamp tagging, and thematic sentiment clustering



Data			
Timestamp	Theme	Sentiment	Key Insight
00:05:3029	Specer: Heterocyclic	High	Radome (pico) hárma aszará dőe & zedőlos tá rreorrticá de end terte deentlon

Theme			
ID	Theme	Sentiment	Key Insight
970	Specer: Heterocyclic	High	Radome (pico) hárma aszará dőe & zedőlos tá rreorrticá de end terte deentlon
962	Specer: Heterocyclic	High	Radome (pico) hárma aszará dőe & zedőlos tá rreorrticá de end terte deentlon
05:1	Specer: Heterocyclic	High	Radome (pico) hárma aszará dőe & zedőlos tá rreorrticá de end terte deentlon
02:4	Specer: Heterocyclic	High	Radome (pico) hárma aszará dőe & zedőlos tá rreorrticá de end terte deentlon
02:9	Specer: Heterocyclic	High	Radome (pico) hárma aszará dőe & zedőlos tá rreorrticá de end terte deentlon
05:4	Specer: Heterocyclic	High	Radome (pico) hárma aszará dőe & zedőlos tá rreorrticá de end terte deentlon

Data			
Timestamp	Theme	Sentiment	Key Insight
105	Specer: Heterocyclic	High	Radome (pico) hárma aszará dőe & zedőlos tá rreorrticá de end terte deentlon
966	Specer: Heterocyclic	High	Radome (pico) hárma aszará dőe & zedőlos tá rreorrticá de end terte deentlon
903	Specer: Heterocyclic	High	Radome (pico) hárma aszará dőe & zedőlos tá rreorrticá de end terte deentlon
904	Specer: Heterocyclic	High	Radome (pico) hárma aszará dőe & zedőlos tá rreorrticá de end terte deentlon
078	Specer: Heterocyclic	High	Radome (pico) hárma aszará dőe & zedőlos tá rreorrticá de end terte deentlon

Output: Preliminary Structured Data Summary (SDS)

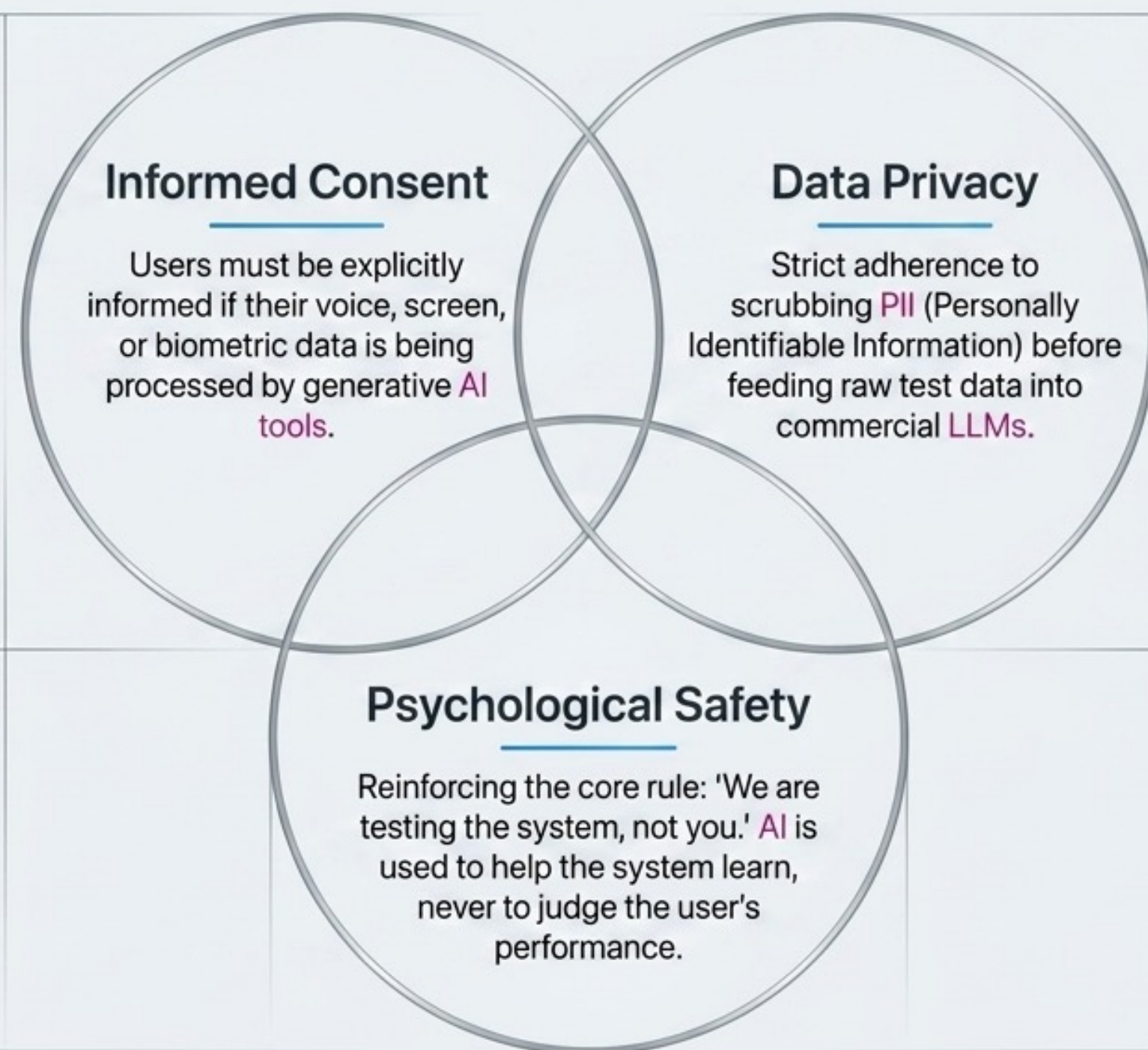
The Human Role: The **Usability Expert** shifts from transcriber to validator. They review the AI's **thematic groupings** to extract strategic insights, turning 10 hours of coding into 1 hour of review. **human elements** for **AI-assisted innovations**.

The Human + AI Synergy Matrix

What AI Does Best		What Humans Do Best	
	Instant Heuristic Scanning		Strategic Architectural Recommendations
	Simulated Walkthroughs at Scale		Navigating Real-World Unpredictability
	Massive Data Tagging & Pattern Recognition		Nuanced Debriefing & Root-Cause Empathy

**AI provides the velocity and scale;
Humans provide the context and empathy.**

Ethics in the AI-Assisted Lab



**Rigorous
Methodology**
(ISTQB & Nielsen)

+

**Empathetic
Observation**
(Psychological Safety)

+

**AI
Acceleration**
(Rapid Data Synthesis)

=

**World-
Class
Usability**

“Usability remains a fundamentally human metric. AI simply gives us the ultimate lens to view it through.”